

KENYA HEALTH FACILITIES ANALYTICS DASHBOARD

Kenya Health Facilities Analytics Dashboard: A Power BI Analysis of Healthcare Infrastructure, Service Availability, Ownership Patterns, and Capacity Distribution

Executive Summary

This dashboard provides a comprehensive analysis of healthcare facilities across Kenya, examining facility distribution, ownership patterns, operational status, service availability, and healthcare capacity indicators. The analysis was developed using Power BI to support evidence-based decision-making in health systems planning, resource allocation, and healthcare service delivery.

Key Findings

The dataset contains approximately 9,000 health facilities distributed across all 47 counties, demonstrating extensive national healthcare coverage. A total bed capacity of approximately 54,000 beds was identified, with an average of 6.09 beds per facility.

Geographic analysis revealed substantial variation in the distribution of health facilities and hospital bed capacity across counties. Certain counties, including Nairobi, Kisumu, Kiambu, Nakuru, and Kakamega, demonstrated significantly higher healthcare capacity and facility concentration compared to others, highlighting regional disparities in healthcare infrastructure.

Healthcare facilities were predominantly concentrated at KePH Level 2, indicating that primary healthcare facilities form the backbone of Kenya's healthcare system. Higher-level facilities, including Levels 4, 5, and 6, represented a smaller proportion of the overall healthcare network but contributed substantially to bed capacity and specialized healthcare services.

Ownership analysis showed that the Ministry of Health remains the largest healthcare provider in Kenya, accounting for nearly half of all facilities. Private enterprises, faith-based organizations, private practitioners, and non-governmental organizations also contribute significantly to healthcare service delivery, demonstrating the importance of public-private collaboration in the health sector.

Service availability analysis indicated that most facilities do not provide 24-hour services or weekend operations, suggesting potential gaps in healthcare accessibility outside regular working hours. These findings highlight opportunities to strengthen emergency and after-hours healthcare coverage, particularly in underserved regions.

Healthcare capacity assessment revealed that higher-level facilities contribute disproportionately to national bed capacity despite representing a smaller share of total

facilities. This emphasizes the critical role of referral hospitals and specialized healthcare institutions in supporting Kenya's health system.

Recommendations

- Strengthen healthcare infrastructure in counties with lower facility density and bed capacity.
- Expand 24-hour and weekend service availability to improve healthcare accessibility.
- Invest in upgrading lower-level facilities to enhance service delivery capacity.
- Support partnerships between public, private, and faith-based healthcare providers.
- Utilize county-level facility and capacity data to guide equitable resource allocation and health workforce planning.

Conclusion

The Kenya Health Facilities Analytics Dashboard provides valuable insights into the structure, distribution, and capacity of the national healthcare system. By integrating facility distribution, ownership, operational status, and capacity indicators into a single analytical platform, the dashboard supports strategic planning and data-driven decision-making aimed at improving healthcare access and health outcomes across Kenya.

Tools Used

- * Power BI
- * DAX Measures
- * Data Modeling
- * Healthcare Facility Data Analysis
- * Data Visualization
- * Public Health Analytics

Prepared by:

Mary Njeri Wanjiku

Data Analyst | Power BI, SQL, DAX | Business Intelligence for Finance & Fintech

Power BI Healthcare Analytics Project, 2026

